Hand and Glove Segmentation Dataset for Department of Energy Glovebox Environments

Dataset Report

Introduction

Many hazardous, industrial tasks require work to be performed in gloveboxes: enclosures that allow workers to manipulate material through gloves without the risks of direct exposure or uncontrolled release of materials. Integrating robots into existing gloveboxes can alleviate common risks associated with glovebox work; e.g., ergonomic injuries and safety risks such as glove tears. However, existing machine learning algorithms that power robotic solutions face challenges regarding their training dataset's diversity, a lack of uncertainty quantification for applications in diverse environments, and a lack of domain relevance to Department of Energy (DOE) glovebox settings. To begin addressing these challenges, a novel dataset featuring pixel-wise labeled hand and glove segmentation masks in a variety of challenging glovebox environments is provided.

The Hand and Glove Segmentation Dataset for Department of Energy (DOE) Glovebox Environments (HAGS) is a robot allocentric perception dataset. It aims to improve safety and accuracy in human-robot collaboration (HRC), particularly in glovebox environments. The dataset's incorporation of diverse HRC experiments, including building a Jenga block tower and disassembling a small box, enhances diversity, variability, and reproducibility, providing a comprehensive representation of interactions for robust and generalizable studies in human-robot interaction. The dataset is instrumental in advancing safety systems by providing challenging prediction scenarios for machine learning algorithms and fostering the development of intelligent, reliable solutions for human-robot collaboration.

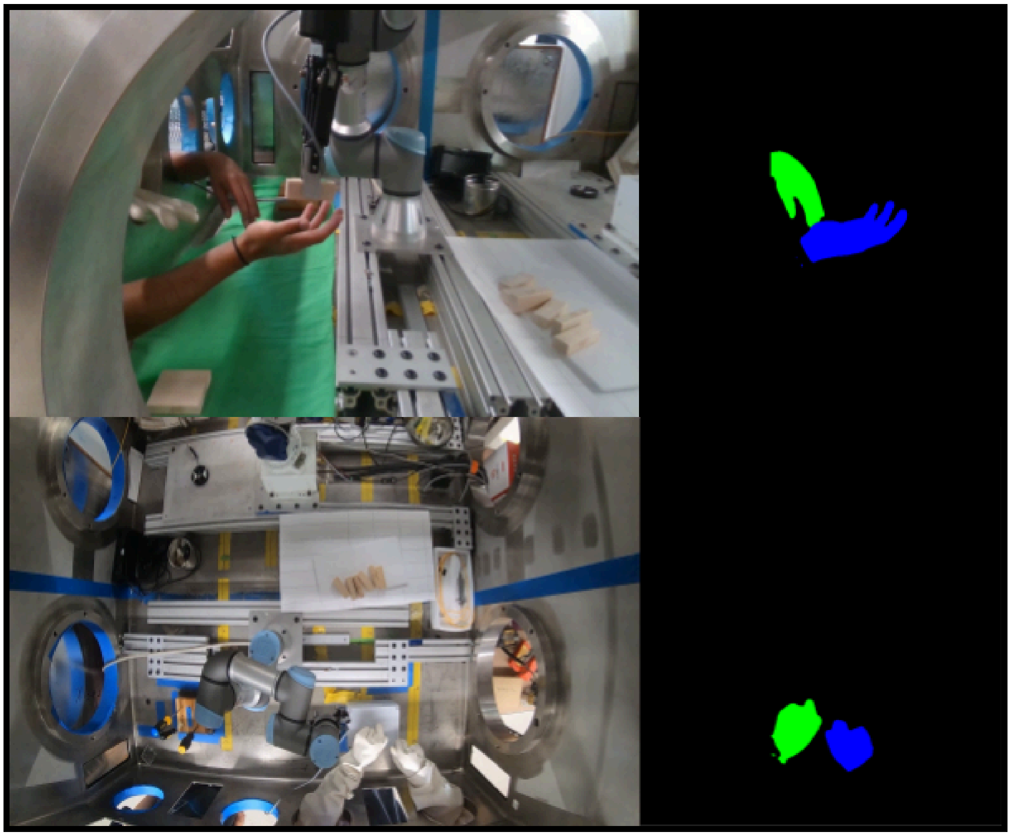

Figure 1: Dataset Preview.

The dataset captures two human-robot collaboration experiments. In the first experiment, participants built a Jenga block tower, receiving six blocks from the robot manipulator arm. The second experiment required participants to disassemble a box, with the robot manipulator arm providing the participant with different screwdrivers to remove four screws. Each participant repeated both experiments four times under the following conditions: a) wearing gloves, b) ungloved, c) with a green screen placed along the bottom of the glovebox, and d) without a green screen placed along the bottom. Lastly, each experiment was recorded from two distinct camera angles: a top view and a side view.

Dataset Contents

The dataset contains:

- Ten participants conducted two experiments, each involving four variables and two camera angles, resulting in a total of 16 videos per participant.
- Eight hours of video footage of each experiment.
- 2876 annotated in-distribution and out-of-distribution frames.
- 1438 original, unannotated sampled frames.

Human Participants

This study was approved by the University of Texas at Austin Institutional Review Board (IRB) under the IRB ID: STUDY00003948. Anyone present in the recorded data and their observed behavior provided consent. To provide a comprehensive representation of collaborative scenarios, a diverse pool of participants was selected. To protect their privacy, participants with recognizable features were asked to cover them up through the use of makeup or other methods. Anyone who revoked their consent and expressed so was noted and removed from the data and the annotations. Included in this data package are the IRB-exempt determination and the Research Information Sheet distributed to the participants.

Robots (Robot Instrumentation)

A Universal Robots UR3e[4] robot manipulator arm equipped with a gripper for object handling was pre-programmed to conduct the two tasks. Additionally, two researchers assisted in the experiment, one operating the robot arm and the other assisting in object placement.

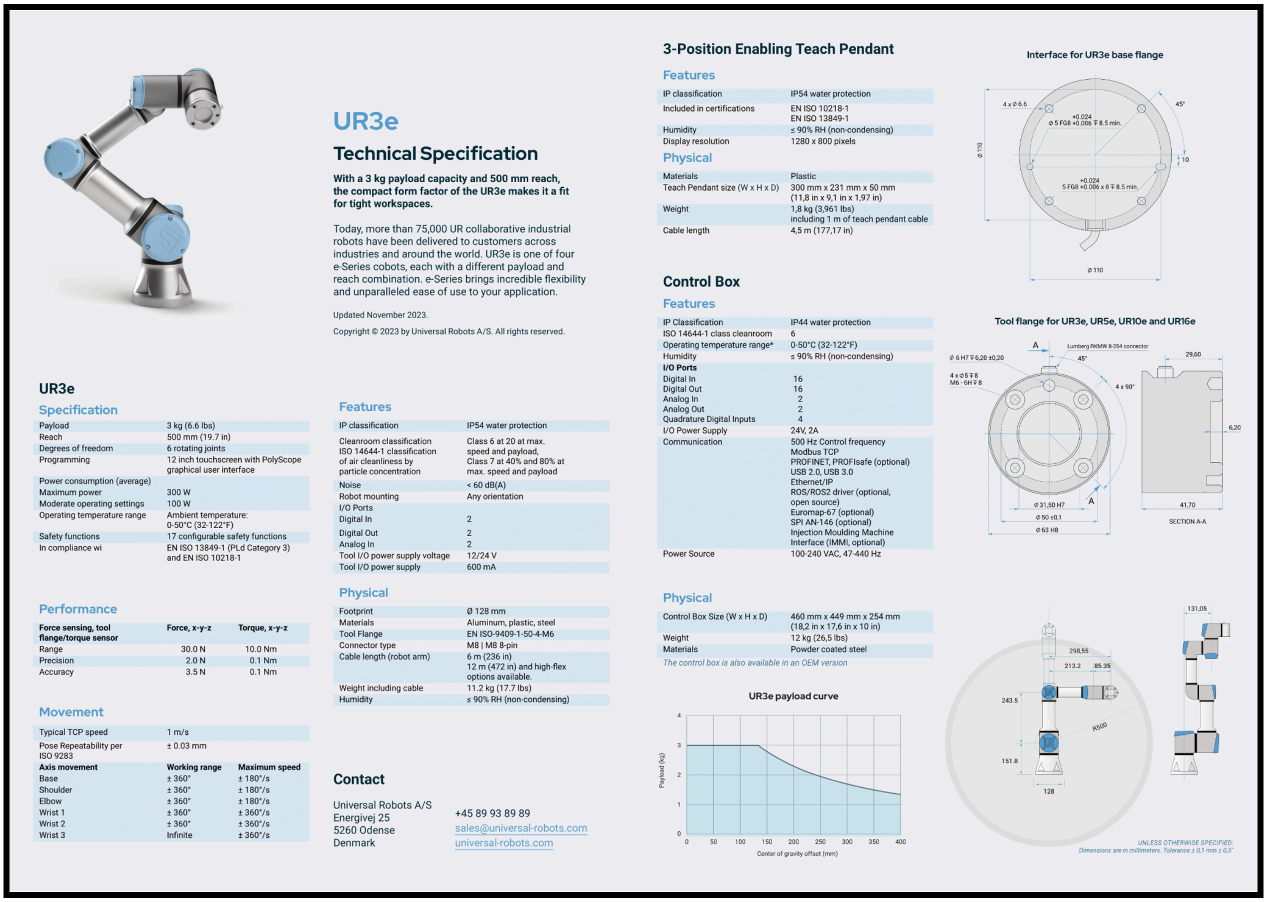

Figure 2: Universal Robots UR3e technical specification.

Experiment Setting

Glovebox

The data was collected in a standard glovebox commonly utilized by researchers in the DOE. The dimensions of the glovebox are 5.04 x 5.04 x 3.71 ft.

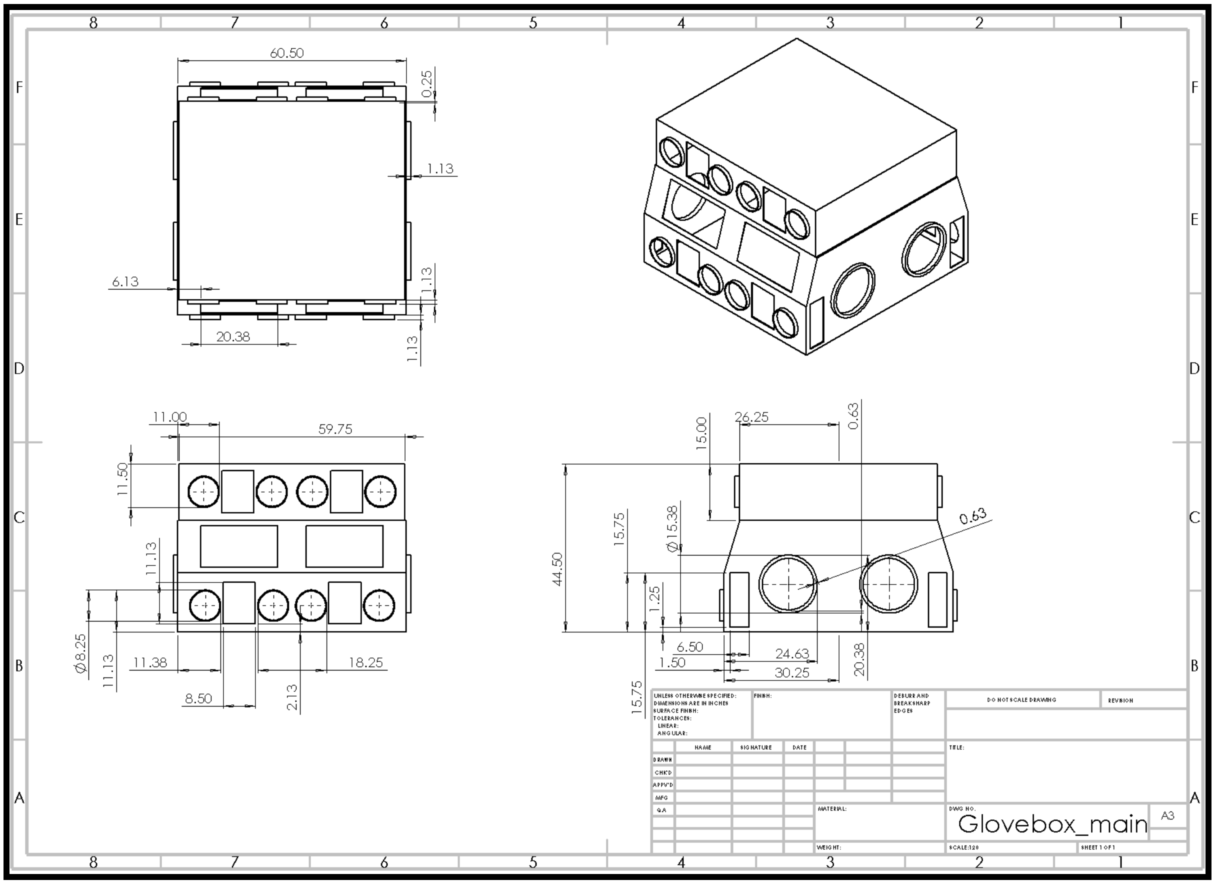

Figure 3: Experimental glovebox technical specification.

Sessions and Tasks

One participant per session completed two tasks, including building a Jenga tower and disassembling a small box. Each task was repeated four times under different conditions: with gloves and a greenscreen, without gloves and a greenscreen, with gloves and no greenscreen, and without gloves and no greenscreen. Overall, each session took around 30 minutes to complete.

Task Descriptions

Two distinct HRC tasks were recorded within this dataset. In the Jenga block tower task, a robot operator facilitates the interaction between one human participant and a block placer. The robot operator approves the robot to grip a block and then moves to provide it to the participant. The gripper opens after approval, allowing the human participant to hold the block. This process repeats until the participant has stacked six blocks in a tower. The use of blocks, resembling human fingers, challenges models to distinguish these factors from real hands, enhancing accuracy in safety systems for scenarios where hands and objects may be perceived as congruent.

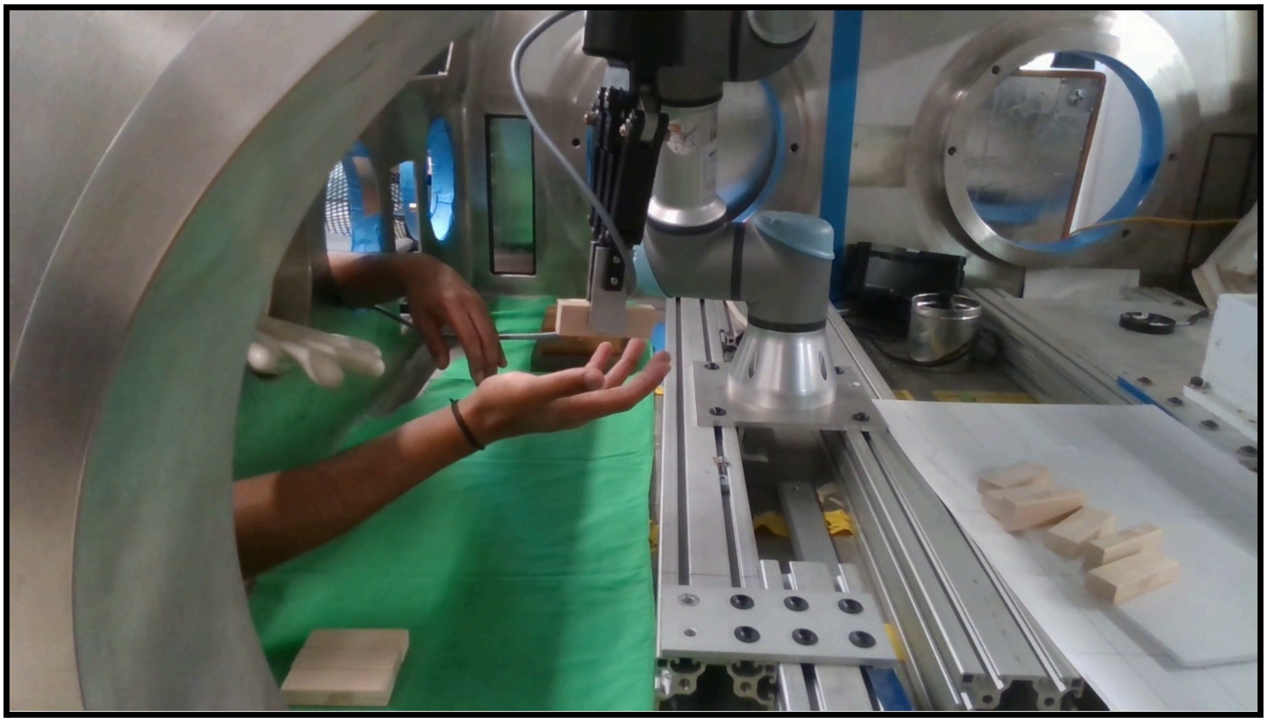

Figure 4: Depicts Jenga block tower task.

For the toolbox task, a human participant is provided with a closed toolbox containing four screws. A tool adjuster is on standby, and a robot operator approves robot moves. The robot iteratively grips tools and provides them to the subject for opening the box. After two tools are used, the participant hands a third tool back to the robot, which then provides it to the designated tool adjuster. The robot picks up the fourth tool and gives it to the human participant. Finally, the robot retrieves the initially rejected tool, providing it to the participant for opening the last screw. Different shapes and sizes of screwdrivers, along with a white toolbox appearance, challenge models to learn physical features to distinguish between the toolbox and the white gloves used by the human participant.

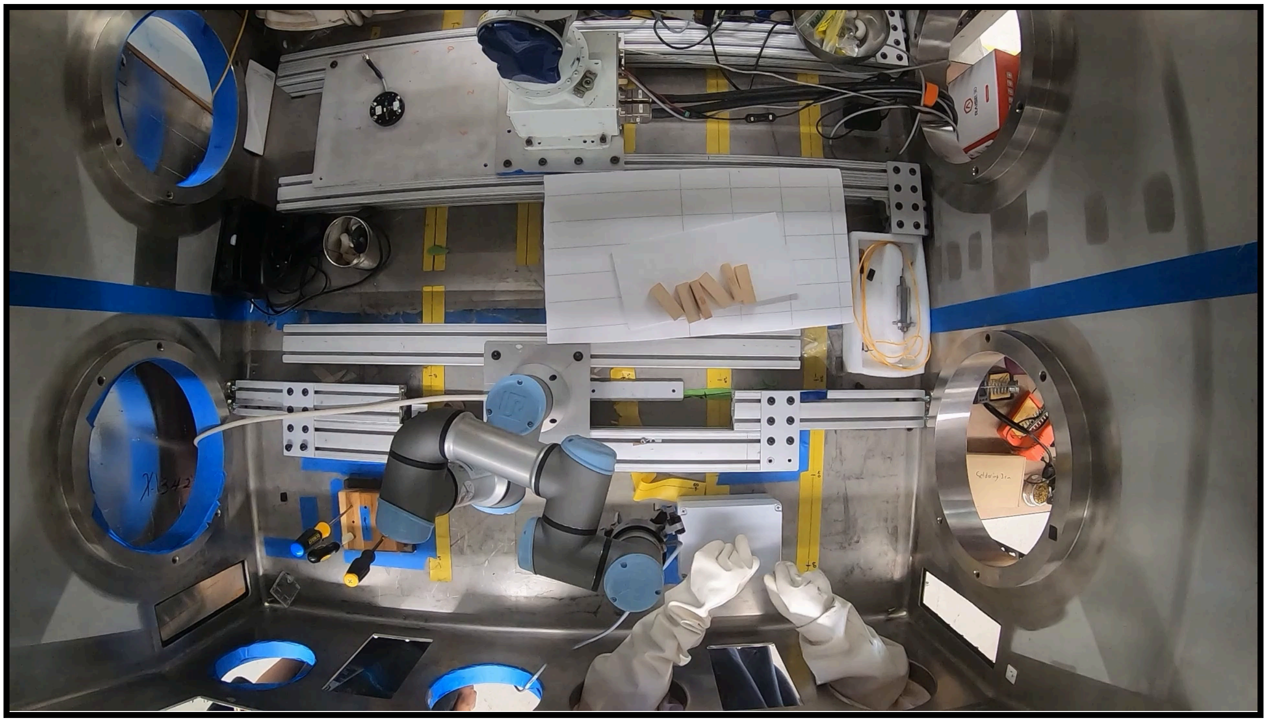

Figure 5: Depicts toolbox task.

Furthermore, in the lab setting, the internal environment of the glovebox varied naturally during usage, with a green screen control measure employed. For half of each participant's videos, a green screen is placed on the bottom of the glovebox, introducing adversarial examples for testing trained models. Frames extracted from videos with a green screen background constitute the out-of-distribution set. Additionally, half of each participant's videos involve situations without gloves, which were also placed in the out-of-distribution set to assess model generalizations in scenarios where gloves may be absent or torn inside the glovebox.

Data acquisition

Each task was captured from two distinct camera angles: one from a bird's eye view captured by a 1080p GoPro Hero 7 Black[1], and another from a 1080p Intel RealSense Development Kit Camera SR300[2] recording from the right side of the participant.

Figure 6: Cameras utilized within the experiment. GoPro Hero 7 Black (left) and Intel RealSense Development Kit Camera SR300 (right).

Dataset

Data Processing

For machine learning applications, the sampled frames underwent a split into two sets: a) the in-distribution set and b) the out-of-distribution set. The in-distribution set encompasses scenarios highly likely in human-robot collaboration within a glovebox, suitable for model training. Consequently, videos without a green screen background and with participants wearing gloves were classified as the in-distribution set. All other videos, featuring participants without gloves and/or a green screen background, less likely in a glovebox setting, were assigned to the out-of-distribution set, facilitating model evaluation. A total of 1438 frames were sampled for labeling, evenly distributed across all videos, with 120 in-distribution frames and 24 out-of-distribution frames per participant (except Participant 6); further explanation regarding the exception for Participant 6 can be found in the Data Quality Statement.

The data was manually annotated by four researchers who were divided into two groups. Each video was cropped to eliminate redundant footage at the beginning and end. Annotation occurred on the sampled frames, with three classes assigned to each image: left hand, right hand,

and background. Human annotators were instructed to annotate each hand from tip to wrist, estimating the wrist location when the subject wore gloves. An open-sourced Segment Anything Model[5] was utilized, alongside the Label Studio[3] annotation tool, initiated with a user box or point prompt. Subsequently, human annotators refined each annotation for precision. To ensure reliability, two annotators labeled each image for inter-annotator agreement. The annotations were converted to single PNG files, depicting the three classes: left hand, right hand, and background, compressed to 64x64 resolution, and generated ground truth masks.

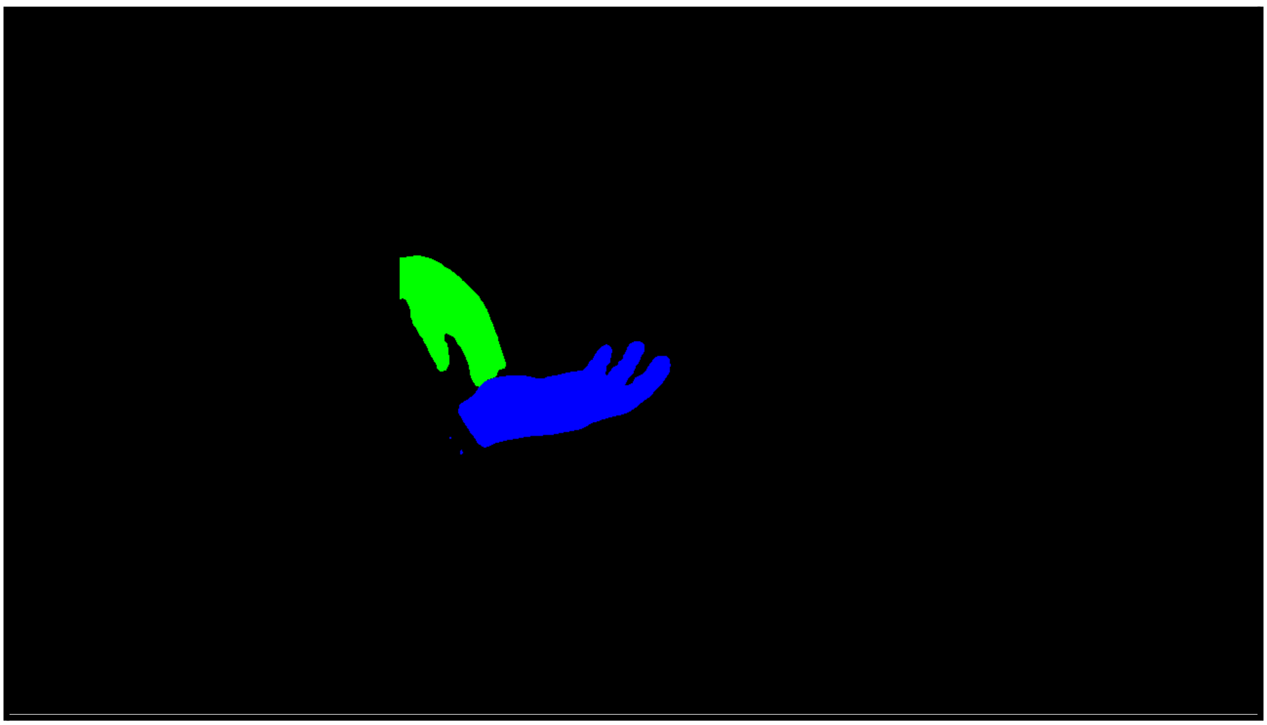

Figure 7: Depicts annotated frame.

Dataset Organization

The dataset is organized as follows:

```
- Dataset
    - Sampled_Frames
        - Test_Subject_{#}
            - ood
```

```
            - {EXPERIMENT}
               - {VIEW}
                  - {EXPERIMENT}_{GLOVE}_{GREENSCREEN}_{#}.png
                  - labelhistory.csv
         - id
            ...
   - Annotations
      - Test_Subject_{#}
         - By_1
            - ood
               - {EXPERIMENT}
                  - {VIEW}
                     - {EXPERIMENT}_{GLOVE}_{GREENSCREEN}_{#}.png
            - id
               ...
         - By_2
            ...
   - Cropped_Videos
      - Test_Subject_{#}
         - Top_View
            - {EXPERIMENT}_{GLOVE}_{GREENSCREEN}.mp4
         - Side_View
            - {EXPERIMENT}_{GLOVE}_{GREENSCREEN}.mp4
```

Three primary categories—`Sampled_Frames`, `Annotations`, and `Cropped_Videos`—comprise the dataset's hierarchical architecture, ensuring a systematic and comprehensive layout.

The first category, `Sampled_Frames`, contains the frames equally sampled from the cropped videos of each participant conducting each task. The subcategory, `Test_Subject_{#}`, distinguishes frames sampled from each test subject. Next, for each test subject, the frames are distinguished as `id`, signifying in-distribution, and `ood`, signifying out-of-distribution. Both `id` and `ood` sublevels are then distinguished by experiment,

`{EXPERIMENT}`, in which there is the option to view the data for 'Jenga_task', the Jenga block tower experiment, and 'Toolbox_task', the toolbox experiment.

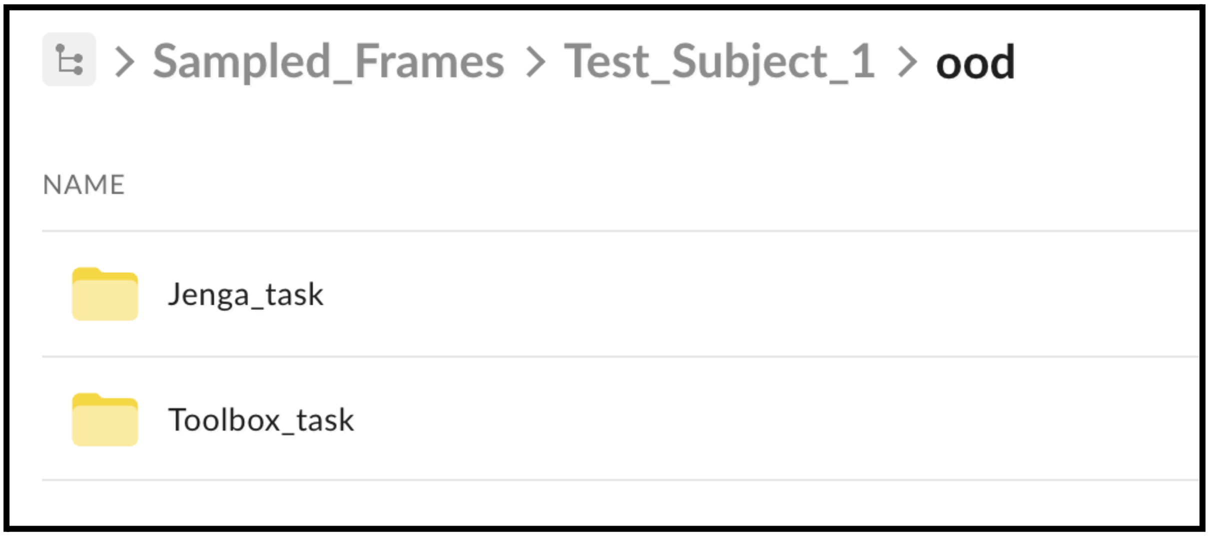

Figure 8: Depicts the `{EXPERIMENT}` level under the `id` and `ood` sub levels.

In the next level, `{VIEW}`, there is the option to select the desired view, being 'Top_View' or 'Side_View'.

Figure 9: Depicts the `{VIEW}` level under the `{EXPERIMENT}` sub-level.

Finally, this brings to the sampled frames, `{EXPERIMENT}_{GLOVE}_{GREENSCREEN}_{#}.png`, labeled based on the experiment

('Toolbox', 'Jenga'), gloves or no gloves ('GL',' NG'), the inclusion of green screen ('G'), and frame number. Furthermore, at this level, there is a CSV file with information about the frames extracted, including timestamps, initial frame number, and extracted frame numbers.

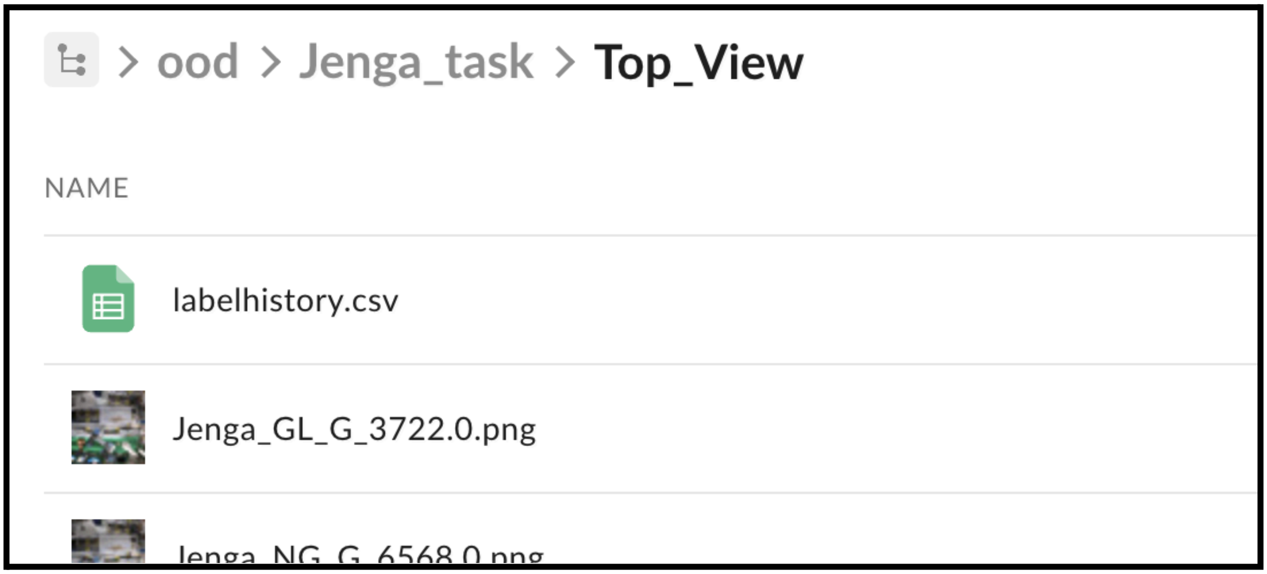

Figure 10: Depicts the `{EXPERIMENT}_{GLOVE}_{GREENSCREEN}_{#}.png` naming convention.

Moving to the second category, `Annotations`, this section provides annotations for the frames sampled in the previous category. The file structure is mostly the same, except for the fact that under `Test_Subject_{#}`, the different annotator groups contributing to the annotations are denoted by `By_1` and `By_2`. Within each annotator group, the `id` and `ood` subcategories maintain a structure analogous to the corresponding sections in `Sampled_Frames`, without the inclusion of a CSV file.

Finally, in the third category, `Cropped_Videos`, the video data of the experimental sessions are included. `Test_Subject_{#}` contains cropped videos for each test subject, categorized into `Top_View` and `Side_View`, each video following a standardized naming convention (`{EXPERIMENT}_{GLOVE}_{GREENSCREEN}.mp4`) specific to the experiment, glove condition, and green screen presence, effectively similar to the convention described in the previous two categories.

Training Models

Using the annotated dataset, multiple semantic segmentation models were trained with the purpose of predicting the hands and generating a label for the images. Results are reported in published papers about this study. The official model training code and configurations for this dataset are in GitHub. The link to the GitHub repository is also provided in the Software metadata field on the Texas Data Repository.

Data Quality Statement

Ensuring the highest quality of data, the research team rigorously implements standardized protocols throughout the experimentation process, ensuring consistency and reproducibility in participant procedures during data collection. Meticulous monitoring of participants verifies adherence to these protocols, thereby maintaining the quality of all data, including videos and pictures captured during the sessions. Throughout the annotation process, inter-annotator agreement is established, with each annotation reviewed by two individuals to enhance accuracy and reliability. Furthermore, comprehensive documentation is meticulously maintained throughout the data collection process, ensuring traceability and facilitating data auditing. All data included in the dataset is thoroughly documented, guaranteeing transparency and reproducibility in the findings.

Three instances of noise are identified in this dataset after quality inspection. Firstly, it is observed that only 22 out-of-distribution frames were sampled with Participant 6, as opposed to the expected 24. This discrepancy occurs specifically within the Side_View of the Jenga_task. Secondly, there is an issue with the orientation of the Top_View > Toolbox_GL video and its subsequent frames and annotations for Participant 3. Finally, the cropped video included in the dataset for Participant 3 > Side_View > Jenga_GL_G is not the same video as the one used for generating the sampled frames and annotations. This leads to a slight inconsistency in terms of reproducibility for this aspect.

References

[1]*Hero7 Black update page - GoPro*. GoPro. (n.d.). https://gopro.com/en/us/update/hero7-black

[2]*Intel® realsenseTM camera SR300 - product specifications*. Intel. (n.d.). https://www.intel.com/content/www/us/en/products/sku/92329/intel-realsense-camera-sr300/specifications.html

[3]Tkachenko, M., Malyuk, M., Holmanyuk, A., & Liubimov, N. (2020). *Label Studio: Data labeling software*. Retrieved from https://github.com/heartexlabs/label-studio

[4]*UR3E - Universal Robots*. UR3e Technical Specification. (2023, November). https://www.universal-robots.com/media/1807464/ur3e-rgb-fact-sheet-landscape-a4.pdf

[5]Zhang, C., Han, D., Zheng, S., Choi, J., Kim, T.-H., & Hong, C. S. (2023, December 15). *Mobilesamv2: Faster segment anything to everything*. arXiv.org. https://arxiv.org/abs/2312.09579